

Isaac Mubiru

isaacmubiru99@gmail.com | linkedin.com/in/isaac-mubiru | github.com/zac-09 | Kampala, Uganda

PROFESSIONAL SUMMARY

Backend engineer with 6+ years building **batch ETL and near-real-time data pipelines** on high-volume production systems. Built a live two-way MongoDB↔Firestore sync on **Google Pub/Sub** that kept 20+ applications consistent under parallel production traffic during a zero-downtime migration; ran scheduled jobs over **2M+ records** to keep customer-facing prices current from external market-data APIs; and architected an **event-driven** microservices backend on **Apache Kafka, Docker and Kubernetes**. Strong **Python**, SQL and NoSQL-at-scale foundations — suited to owning data pipelines, API integrations and data quality for ML and analytics teams.

CORE COMPETENCIES

Batch ETL & Data Pipelines	Near-Real-Time Data Sync	Event-Driven Architecture	Data Integration & Migration	Python & SQL
NoSQL at Scale (MongoDB, Firestore)	Apache Kafka & Pub/Sub	Docker & Kubernetes		

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer · US Remote

- Built a **near-real-time two-way data pipeline** between MongoDB and Firestore using **Node.js** and Google **Pub/Sub** for message processing, keeping both stores consistent while serving live production traffic.
- Wrote a **Python** pipeline to migrate file storage from Amazon **S3** to Google Cloud Storage as part of a wider cloud consolidation.
- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Shipped a 3D visualisation feature (video overlay + ffmpeg) and Express Shipping (+\$40 revenue/order), improving buyer conversion; cut infrastructure cost by **\$5,000/month** through cloud consolidation.

Dr Wealth Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Built scheduled jobs querying **over 2 million Firestore records** to keep customer-facing stock prices current from **Morningstar APIs** — high-volume ingestion against an external data source with a strict freshness requirement.
- Extended backend APIs with Firebase Cloud Functions and **Express.js** on Heroku; added responsive PWA pages in React and Tailwind.

MTailor Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Built the **ETL pipeline** that extracted from MongoDB, applied complex transformation logic, and loaded into Firestore for a full backend migration.
- Documented the new Firebase SDKs (**Node.js and Python**) for engineers and trained the Ops team on dashboards and backup procedures.

CodeBits Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- Architected an **event-driven microservices** backend for FIDA Uganda's case-management platform using **Apache Kafka, Docker and Kubernetes**, with NATS Streaming for service messaging.
- Built a USSD legal-aid service integrating with a **Node.js** backend via **gRPC**; led a team of 4 developers.
- Delivered a paralegal database for LASPNET Uganda and a cross-platform React Native app with Firebase push notifications.

Mind2matter Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using **Node.js** and Web3; delivered React UIs from Figma under tight timelines.

PROJECTS

Real-time Pub/Sub Sync Pipeline

Two-way MongoDB↔Firestore data pipeline with idempotent, redelivery-safe writes, running under parallel production traffic across 20+ applications during a zero-downtime migration.

Node.js · Google Pub/Sub · MongoDB · Firestore

High-Volume Market Data Refresh

Scheduled jobs keeping 2M+ Firestore records current from external Morningstar market-data APIs for a live customer-facing pricing PWA.

Node.js · Firestore · External API ingestion

Event-Driven Microservices — FIDA Uganda

Architected a Kafka-based event-driven backend for a legal case-management platform, containerised and orchestrated on Kubernetes.

Apache Kafka · Docker · Kubernetes · gRPC · NATS Streaming

Cross-Cloud Storage Migration

Python batch pipeline migrating object storage from Amazon S3 to Google Cloud Storage with integrity verification.

Python · AWS S3 · Google Cloud Storage

EDUCATION

B.Sc. Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

LINKS

Open-source & project code — github.com/zac-09

SKILLS

Data & Pipelines: Batch ETL, Near-real-time pipelines, Apache Kafka, Google Pub/Sub, NATS Streaming, Data integration & migration

Databases: MongoDB, Firestore, SQL, Data modelling (document stores)

Languages: Python, JavaScript, TypeScript, SQL, Go (intermediate), Rust (intermediate)

Cloud / Infra: GCP (Firestore, Pub/Sub, Cloud Functions, GCS), AWS (S3, EC2, EBS — intermediate), Docker, Kubernetes, Linux

Backend: Node.js, Express.js, REST APIs, gRPC, Microservices **Spoken:** English, Kiswahili